

# Technical documentation

## WEB API Framework

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# 1 Introduction

This document contains all the information regarding the implementation of the technical aspects of the Web API framework.

## 2 Example usage

This chapter contains examples on how to build a REST service using the library. It explains what you need to start, how concepts like routers work, how you make your own endpoint and configure it, how you can build a login endpoint, how you can add extra functionality to your REST service using modifiers and how to enable authentication.

### 2.1 Building a very basic REST service

For this example we'll take the WebOrder workspace that is installed along with the DataFlex installation to build a very basic REST service.

To start open the WebOrder workspace in your studio. Preferably the 25.0 studio as this version comes with features that allow drag and drop to work from the DDO explorer.

In the studio toolbar click Tools -> Maintain libraries. The Library maintenance panel should now pop up. Navigate to the folder where you installed the library. Select the .sws version that matches your DataFlex version.

Now that the library is in the workspace you can start creating a REST service. The easiest way to get started is to the Web HTTP handler wizard. In the studio toolbar select File -> New -> Web Object -> Web HTTP Handler. The wizard pops up and asks for a name. For this example I'll call it oMyRestAPI. This file should now be open in the studio.

**Use** cWebHttpHandler.pkg

```
// With the cWebHttpHandler you handle complete HTTP requests.
```

**Object** oMyRestAPI **is a** cWebHttpHandler

```
// The psPath property determines the path in the URL for which requests will be handled.
```

```
Set psPath to "MyHandler"
```

```
// Use psVerbs to filter based on the HTTP Verbs.
```

```
Set psVerbs to "*"
```

**End\_Object**

You can get rid of all the current code inside of the cWebHttpHandler. After this change the use statement to "Use cWebApi.pkg" and change the class of the object from cWebHttpHandler to cWebApi. Your file should now look something like this:

**Use** WebApi\cWebApi.pkg

**Object** oMyRestAPI **is a** cWebApi

**End\_Object**

If you now navigate to the WebOrder url and append /Api to the url you'll send a request to this object. So for DataFlex 25.0 I would send a request to: [http://localhost/WebOrder\\_25\\_0/Api](http://localhost/WebOrder_25_0/Api) . If you already have a existing WebHttpHandler that listens to /Api you can change it by setting the psPath of the cWebApi object. For example if you set the psPath to "RestLibraryService" you would need to send a request to: [http://localhost/WebOrder\\_25\\_0/RestLibraryService](http://localhost/WebOrder_25_0/RestLibraryService) .

You'll see that if you send a request to this REST service now you receive no response. This is because there is no content inside of our REST service.

Before we add any content we need to tell the cWebApi object what datatypes we want to support for our REST service. For example do we want JSON, XML or maybe we want both? To do this we use the AddIterator command inside of the cWebApi object. This procedure takes two parameters. The first being a reference to the class that it needs to use to build up the response in that specific data type. The second parameter is what accept-type/content-type it will be used for. Keep in mind that the first iterator you register to the cWebApi object will be the default. So if someone sends a request with a not supported accept-type or content-type it will default to the first iterator. This will look something like the following:

```
Object oMyRestAPI is a cWebApi
    Send AddIterator (RefClass(cJSONIterator))
    "application/json"
    Send AddIterator (RefClass(cXMLIterator)) "application/xml"
End_Object
```

This basically means that whenever we receive a request with accept-type or content-type "application/json" we'll use the cJSONIterator class to handle this request. And the same goes for the cXMLIterator whenever we receive a request with "application/xml". And as previously mentioned, in this case the default is the cJSONIterator.

Now that the REST service knows what datatypes are supported. We can start adding our data. To make our cWebApi less cluttered we will create our endpoint in a separate file.

In the toolbar select File -> New -> Other -> DataFlex Source File. For this example I'll be calling this file "MyFirstEndpoint.pkg". For this example lets expose the inventory table in the WebOrder database.

To start import the cRestDataset class by typing "Use cRestDataset.pkg". The cRestDataset is the basis for your data aware endpoints. Now make a object of the cRestDataset class. I'll be calling it oMyEndpoint in this example. Set the psPath property of this object to Inventory. Your code should now look something like this:

```
Use WebApi\cRestDataset.pkg
```

```
Object oMyEndpoint is a cRestDataset
    Set psPath to "Inventory"
End_Object
```

Go back to the cWebApi object and add the use statement to the new file inside of it. This should look like the following:

```
Object oMyRestAPI is a cWebApi
    Send AddIterator (RefClass(cJSONIterator))
    "application/json"
    Send AddIterator (RefClass(cXMLIterator)) "application/xml"

    Use MyFirstEndpoint.pkg
End_Object
```

Our endpoint is now registered in the cWebApi object. This means that if we now send a request to [http://localhost/WebOrder\\_25\\_0/Api/Inventory](http://localhost/WebOrder_25_0/Api/Inventory). It will reach our endpoint. However as of right now it has no content. To change this lets add the needed data dictionaries. Inside of your endpoint open the DDO explorer, select "Add DDO" and select the cInventoryDataDictionary. Your endpoint should now look something like this:

```
Object oMyEndpoint is a cRestDataset
    Set psPath to "Inventory"

    Object oVendor_DD is a cVendorDataDictionary
    End_Object

    Object oInventory_DD is a cInventoryDataDictionary
        Set DDO_Server to oVendor_DD
    End_Object

    Set Main_DD to oInventory_DD
    Set Server to oInventory_DD

End_Object
```

If you now send a request to this endpoint you'll see that you get a bunch of empty object. This is because we have not defined the data we want to return yet. For this example lets expose the following fields from the inventory table:

- Item\_ID
- Description
- Unit\_Price
- On\_Hand

To do this we'll use the cRestField. This is what allows you to expose your fields. The cRestField has some options you can configure. For example you can make a field ReadOnly by setting the property pbReadOnly to true. This means that the framework will allow this field if its present in the request body for a POST, PUT or PATCH request. You can also make a field WriteOnly, this means that you can send the field to the server through a request body, but it will not be returned in the response. This is especially usually for sensitive data for password fields, or when a user makes some form of login attempt. You can filter on all defined fields by default. This can be done by adding a query parameter to your url. For example if I wanted all inventory records that have "BEARS" as their Item Id I could use the following url:

[http://localhost/WebOrder\\_25\\_0/Api/Inventory?Item%20Id=BEARS](http://localhost/WebOrder_25_0/Api/Inventory?Item%20Id=BEARS) . This will give me all the records where Item Id is "BEARS". You can add multiple query parameters by using "&". If you do not want the field to be filterable you can set pbFilterable to false.

By default the cRestField will have the same name as the field in your data dictionary. If you wish to give this a custom name set the psFieldName property.

If you are following this guide in DF 25 you can select the above fields in the DDO explorer and drag them inside of your endpoint. The cRestFields will automatically be created with the correct entry\_item. If you are not using this library in DF 25 you will need to create these objects manually and add the correct entry\_item.

After you've added these cRestFields your code should look something like the following:

```
Object oMyEndpoint is a cRestDataset
  Set psPath to "Inventory"

Object oVendor_DD is a cVendorDataDictionary
End_Object

Object oInventory_DD is a cInventoryDataDictionary
  Set DDO_Server to oVendor_DD
End_Object

Set Main_DD to oInventory_DD
Set Server to oInventory_DD

Object oInventory_Item_ID is a cRestField
  Entry_Item Inventory.Item_ID
End_Object

Object oInventory_Description is a cRestField
  Entry_Item Inventory.Description
End_Object

Object oInventory_Unit_Price is a cRestField
  Entry_Item Inventory.Unit_Price
End_Object

Object oInventory_On_Hand is a cRestField
  Entry_Item Inventory.On_Hand
End_Object

End_Object
```

If you now send a request to this endpoint you'll see that each object now contains the fields we have defined above. This is how you setup a very basic endpoint. This automatically enables the GET, POST, PUT, PATCH and

DELETE verbs. If you want to disable some of these you can take a look at the pbAllowRead, pbAllowCreate, pbAllowEdit and pbAllowDelete properties of the cRestDataset object.

All data dictionary operations follow the business rules defined in your data dictionary. So if there is a required field that is not exposed by your endpoint you will not be able to create new records as validation will fail.

If you've followed the steps above you now have configured your first very basic REST service. The framework will automatically create a OpenApi specification based on the endpoints you created. The OpenApi specification is a description of your REST service and can be used in tools like SwaggerUI to create a documentation page. Explanation on how to incorporate SwaggerUI into your application is explained in a later chapter.

## 2.2 Adding information from a parent table to a endpoint

cRestFields are used to expose data from a table. However if we want to add data from a external table its nice to have a visual difference between the data from the main table and the parent table. In JSON this could be a nested JSON object that has all the information related to the parent table. To get this working you'll need to use the cRestEntity class. This class lets the framework know that we're using data from a different table.

The first step would be to add a object that is a cRestEntity. Its important that you set the server of this object to the data dictionary that manages this parent table. For this example we'll add the Vendor table to the inventory endpoint. The endpoint will now look something like the following:

```
Object oMyEndpoint is a cRestDataset
  Set psPath to "Inventory"

  Object oVendor_DD is a cVendorDataDictionary
  End_Object

  Object oInventory_DD is a cInventoryDataDictionary
    Set DDO_Server to oVendor_DD
  End_Object

  Set Main_DD to oInventory_DD
  Set Server to oInventory_DD

  Object oInventory_Item_ID is a cRestField
    Entry_Item Inventory.Item_ID
  End_Object

  Object oInventory_Description is a cRestField
    Entry_Item Inventory.Description
  End_Object

  Object oInventory_Unit_Price is a cRestField
    Entry_Item Inventory.Unit_Price
  End_Object

  Object oInventory_On_Hand is a cRestField
    Entry_Item Inventory.On_Hand
  End_Object

  Object oVendorEntity is a cRestEntity
    Set Server to oVendor_DD

End_Object

End_Object
```

If we were to send a request to this endpoint now you'll see that each record we receive also has a nested "Vendor" object that is empty. To add data to this object we can use cRestFields again, just like how we did it with the cRestDataset. For this example I'll add the following fields to the vendor entity:

- Name
- Address
- City
- State

The entity object should now look something like this:

```
Object oVendorEntity is a cRestEntity
  Set Server to oVendor_DD

  Object oVendor_Name is a cRestField
    Entry_Item Vendor.Name
  End_Object

  Object oVendor_Address is a cRestField
    Entry_Item Vendor.Address
  End_Object

  Object oVendor_City is a cRestField
    Entry_Item Vendor.City
  End_Object

  Object oVendor_State is a cRestField
    Entry_Item Vendor.State
  End_Object

End_Object
```

If you now send a request to the endpoint you'll see that each vendor object is filled with the defined information. By default the entity's name will be whatever your table name is. If you want to change this to something custom you can use the psNodeName property.

## 2.3 Adding information from a child table to an endpoint

cRestFields are used to expose data from the main table. However if we want to add data from a child table its nice to have a visual difference between the data in the response. In JSON this would be a nested array of objects. The framework also somehow needs to know that it needs to find multiple related records. To do this you can use the cRestChildCollection class.

For this example we'll make an endpoint that exposes the vendor table. For each vendor record we'll show all related inventory records. This is the basis we'll start with:

```
Use WebApi\cRestDataset.pkg
Use cVendorDataDictionary.dd
Use cInventoryDataDictionary.dd
Use WebApi\cRestField.pkg

Object oVendorInventory is a cRestDataset
  Set psPath to "VendorInventory"

  Object oVendor_DD is a cVendorDataDictionary
  End_Object

  Object oInventory_DD is a cInventoryDataDictionary
    Set Constrain_file to Vendor.File_number
    Set DDO_Server to oVendor_DD
  End_Object

  Set Main_DD to oVendor_DD
  Set Server to oVendor_DD

  Object oVendor_ID is a cRestField
    Entry_Item Vendor.ID
    Set pbReadOnly to True
  End_Object

  Object oVendor_Name is a cRestField
    Entry_Item Vendor.Name
  End_Object

  Object oVendor_Address is a cRestField
    Entry_Item Vendor.Address
  End_Object

  Object oVendor_City is a cRestField
    Entry_Item Vendor.City
  End_Object

  Object oVendor_State is a cRestField
    Entry_Item Vendor.State
  End_Object

End_Object
```

To add the information from a child table to this we'll make an object that is a `cRestChildCollection` and set its server to the inventory data dictionary. This looks like the following:

```
Object oInventoryCollection is a cRestChildCollection
    Set Server to oInventory_DD
```

```
End_Object
```

We can now start adding `cRestFields` to it to define the data. For this example we'll add the following data:

- Item\_ID
- Description
- Unit\_Price
- On\_Hand

This should now look something like this:

```
Object oInventoryCollection is a cRestChildCollection
    Set Server to oInventory_DD
```

```
    Object oInventory_Item_ID is a cRestField
        Entry_Item Inventory.Item_ID
    End_Object
```

```
    Object oInventory_Description is a cRestField
        Entry_Item Inventory.Description
    End_Object
```

```
    Object oInventory_Unit_Price is a cRestField
        Entry_Item Inventory.Unit_Price
    End_Object
```

```
    Object oInventory_On_Hand is a cRestField
        Entry_Item Inventory.On_Hand
    End_Object
```

```
End_Object
```

If you now send a request to [http://localhost/WebOrder\\_25\\_0/Api/VendorInventory](http://localhost/WebOrder_25_0/Api/VendorInventory) . You'll see that you now get all the related inventory records along with the vendor information. By default it will take the table name. However if you wish to change this you can set the `psNodeName` property.

## 2.4 How to use routers

Objects of the cWebApiRouter class are a extra step in the routing functionality of the framework. This class can be especially useful when you want to add something like versioning to your API or if you want a cWebApiModifier/cWebApiAuthModifier (what these are is explained in a later chapter) to only work for a select few endpoints.

To define a router you create a object of the cWebApiRouter class. A router only has two properties. The one that is relevant for this guide is the psPath property. The pbInheritSecurity property will be explained in the chapter about cWebApiAuthModifiers.

You can define endpoints inside of a router just like how you can define them in the cWebApi object. Endpoints defined inside of a cWebApiRouter have the psPath of the router appended to a url. Take this REST service for example:

```
Object oMyRestAPI is a cWebApi
  Send AddIterator (RefClass(cJSONIterator))
  "application/json"
  Send AddIterator (RefClass(cXMLIterator)) "application/xml"

Object oMyRouter is a cWebApiRouter
  Set psPath to "v1"

End_Object

Use MyFirstEndpoint.pkg
Use VendorInventory.pkg
End_Object
```

Right now if I want to reach the endpoint we created in the very basic REST service guide I would need to send a request to [http://localhost/WebOrder\\_25\\_0/Api/Inventory](http://localhost/WebOrder_25_0/Api/Inventory). However if I put the endpoint inside of the router like this:

```
Object oMyRestAPI is a cWebApi
  Send AddIterator (RefClass(cJSONIterator))
  "application/json"
  Send AddIterator (RefClass(cXMLIterator)) "application/xml"

Object oMyRouter is a cWebApiRouter
  Set psPath to "v1"

  Use MyFirstEndpoint.pkg
End_Object

Use VendorInventory.pkg
End_Object
```

It would be reachable at: [http://localhost/WebOrder\\_25\\_0/Api/v1/Inventory](http://localhost/WebOrder_25_0/Api/v1/Inventory). This technique can be useful if you want multiple versions of the same endpoint.

It is also possible to define cWebApiModifiers and cWebApiAuthModifiers inside of these routers. This way the modifiers functionality only applies to requests that pass through that specific router. This allows you to for example, only apply authentication for a certain set of endpoints.

It is also possible to nest routers inside of eachother. This can be useful if you have a set of public and private routers. This could look something like the following:

```
Object oMyRestAPI is a cWebApi
    Send AddIterator (RefClass(cJSONIterator))
    "application/json"
    Send AddIterator (RefClass(cXMLIterator)) "application/xml"

Object oMyRouter is a cWebApiRouter
    Set psPath to "v1"

    Object oPrivateRouter is a cWebApiRouter
        Set psPath to "private"

        Use MyFirstEndpoint.pkg
    End_Object

    Object oPublicRouter is a cWebApiRouter
        Set psPath to "public"
    End_Object

End_Object

Use VendorInventory.pkg
End_Object
```

The endpoint is now reachable at [http://localhost/WebOrder\\_25\\_0/Api/v1/private/Inventory](http://localhost/WebOrder_25_0/Api/v1/private/Inventory).

## 2.5 Implementing modifiers

Modifiers are extra features you want to add to your REST service such as logging. You build a modifier by using the `cWebApiModifier` class. There are types of modifiers. The one this chapter focuses on is the `cWebApiModifier` object, this class is meant for implementing features such as logging. The second type is the `cWebApiAuthModifier` which enables you to implement authentication/authorization. That class will be explained in the next chapter.

The `cWebApiModifier` has two types of events. The first being the `OnPreRequest` event and the second being the `OnPostRequest` event. Both of these events get a `webapicalcontext` passed as a parameter. This `webapicalcontext` struct is filled with information related to the current request.

The framework will first determine what endpoint the call is being sent to. The framework will remember all the modifiers defined inside of the `cWebApi`, `cWebApiRouter` and endpoint classes that it needed to navigate to the endpoint.

After this, right before the framework starts formulating a response to the client it will call the `OnPreRequest` event of all the modifiers that it found for the current request. For a logging object this is a good place to log information such as:

- The verb used in the current request
- The time the request was received
- The path of the endpoint

Most of this information can be found inside of the `webapicalcontext` struct.

Right before the message is returned to the client the `OnPostRequest` event is called. For a logging class this is a good place to find information such as:

- The time it needed to handle the request
- The status code of the request

Finally you can save the information to some table or maybe just write it to a file. The following example logging class is also included in the example workspace:

```
Object oRequestLogger is a cWebApiModifier

Object oApiLogs_DD is a cApiLogsDataDictionary
End_Object

//Logs the incoming request time
Procedure OnPreRequest tWebApiCallContext ByRef
webapicalcontext
    String sIncomingTime

    Send Clear of oApiLogs_DD

    Move (CurrentDateTime()) to sIncomingTime

    Set Field_Changed_Value of oApiLogs_DD Field
ApiLogs.Verb to webapicalcontext.sVerb
    Set Field_Changed_Value of oApiLogs_DD Field
ApiLogs.TimeReceived to sIncomingTime
    Set Field_Changed_Value of oApiLogs_DD Field
ApiLogs.Endpoint to webapicalcontext.sPath
End_Procedure

//Logs outgoing request time, status code and saves the
record
Procedure OnPostRequest tWebApiCallContext ByRef
webapicalcontext
    Boolean bErr
    String sResponseTime

    Move (CurrentDateTime()) to sResponseTime

    Set Field_Changed_Value of oApiLogs_DD Field
ApiLogs.TimeResponse to sResponseTime
    Set Field_Changed_Value of oApiLogs_DD Field
ApiLogs.StatusCode to webapicalcontext.iStatusCode

    Get Request_Validate of oApiLogs_DD to bErr

    If (not(bErr)) Begin
        Send Request_Save of oApiLogs_DD
    End
End_Procedure

End_Object
```

## 2.6 Implementing authentication/authorization

Authentication can be built into the framework using the `cWebApiAuthModifier` class. This class has three events that you can augment. Those being the `OnPreRequest`, `OnPostRequest` and `OnAuth` events. The `OnPreRequest` and `OnPostRequest` are not that important for the `cWebApiAuthModifier` but they are there if you wish to use them. If you do wish to augment the `OnPreRequest` make sure you do a forward send or the main functionality of this class will not work.

The most important part of this class is the `OnAuth` event. This event is fired whenever this modifier is part of the current request and the endpoint used in the call is secured by this `cWebApiAuthModifier`.

By default the `cWebApiAuthModifier` secures all endpoints defined on the same level as this modifier and each endpoint defined on a lower level. For example if a `cWebApiAuthModifier` is defined directly inside of the `cWebApi` object like so:

```
Object oMyRestAPI is a cWebApi
  Send AddIterator (RefClass(cJSONIterator))
  "application/json"
  Send AddIterator (RefClass(cXMLIterator)) "application/xml"

Object oMyAuthModifier is a cWebApiAuthModifier
End_Object

Object oMyRouter is a cWebApiRouter
  Set psPath to "v1"

  Use MyFirstEndpoint.pkg
End_Object

Use VendorInventory.pkg
End_Object
```

In this example the `OnAuth` event will be called when a request is sent to either the endpoint defined in `MyFirstEndpoint.pkg` or the one defined in `VendorInventory.pkg`. However you were to place the `oMyAuthModifier` object inside of the `oMyRouter` object like so:

```
Object oMyRestAPI is a cWebApi
  Send AddIterator (RefClass(cJSONIterator))
  "application/json"
  Send AddIterator (RefClass(cXMLIterator)) "application/xml"

Object oMyRouter is a cWebApiRouter
  Set psPath to "v1"

  Object oMyAuthModifier is a cWebApiAuthModifier
  End_Object

  Use MyFirstEndpoint.pkg
End_Object

Use VendorInventory.pkg
End_Object
```

The OnAuth event will only be called when a request is being sent to the endpoint defined inside of "MyFirstEndpoint.pkg". Both the dataset classes and the cWebApiRouter classes have a property called pbInheritSecurity. Whenever this is set to false it will not use the cWebApiAuthModifiers defined on a higher level. If you take the first example where the oMyAuthModifier object is defined inside of the cWebApi object, if you were to set pbInheritSecurity of the oMyRouter object like so:

```
Object oMyRestAPI is a cWebApi
    Send AddIterator (RefClass(cJSONIterator))
    "application/json"
    Send AddIterator (RefClass(cXMLIterator)) "application/xml"

Object oMyAuthModifier is a cWebApiAuthModifier
End_Object

Object oMyRouter is a cWebApiRouter
    Set psPath to "v1"
    Set pbInheritSecurity to False

    Use MyFirstEndpoint.pkg
End_Object

Use VendorInventory.pkg
End_Object
```

The OnAuth event of the oMyAuthModifier object will not be fired when a request is being sent towards any endpoint defined inside of the oMyRouter object. This allows you more control over what endpoints you really want to secure.

The OnAuth event has a single parameter, this is the webapicallcontext struct. This is a important parameter that allows us to define if the current request should have access to our current resource.

The webapicallcontext struct has a bunch of members. The members that are most important for this event are the following members:

- bErr
- iStatusCode
- sShortStatusMessage
- sErrorMessage

Whenever bErr is set to true the framework will know that it should not proceed with the current request and will attempt to build up a error response to the client. It will do so using the iStatusCode, sShortStatusMessage and sErrorMessage.

The message will be formatted according to the following RFC: <https://www.rfc-editor.org/rfc/rfc9457.html>

iStatusCode defines the status code returned, sShortStatusMessage is the text that accompanies the status code, for a 403 this would be "forbidden" for example. And sErrorMessage can be a longer custom string to return a more detailed error explanation.

So you can perform any logic you want in terms of authentication, when the condition you set is not met you set `bErr` to true and configure the other members and the framework will handle the rest automatically. The sample workspace has a few example authentication/authorization mechanisms. The most basic one is the `BasicAuth` authentication mechanism which sends a base64 encoded username and password in the `Authorization` header of the request.

## 2.6.1 Including authentication/authorization into the OpenApi specification

In order to reflect your authentication/authorization mechanism into the OpenApi specification two steps should be taken.

The first step that should be taken is to set the property `psSecuritySchemaName` to the name of your security schema. You can find these names on the following page:

[https://swagger.io/docs/specification/v3\\_0/authentication/](https://swagger.io/docs/specification/v3_0/authentication/)

So for example if you were to implement username and password authentication using `BasicAuth` you would set the `psSecuritySchemaName` to `basicAuth`.

The second step is to augment the `OnDefineAuthSchema` event. This event has one parameter which is called `vSecurityStruct`. You should move a struct inside of this variant that matches the schema described in the swagger documentation. For `basicAuth` the schema looks like the following:

```
basicAuth: # <-- arbitrary name for the security scheme
  type: http
  scheme: basic
```

We already defined `basicAuth` as the `psSecuritySchemaName`. So all that is left is to create a struct that has the following two members:

- Type
- Scheme

The value of `type` should be `http` and the value of `scheme` should be `basic`. Our code would look something like the following:

```
Struct basicAuth
  String type
  String scheme
End_Struct

Procedure OnDefineAuthRules Variant ByRef vSecurityStruct
  basicAuth basicauth

  Forward Send OnDefineAuthRules (&vSecurityStruct)

  Move "http" to basicauth.type
  Move "basic" to basicauth.scheme

  Move basicauth to vSecurityStruct
End_Procedure
```

The OpenApi specification should now properly reflect the authentication/authorization mechanism.

## 2.7 Building custom endpoints

The cRestDataset aims to streamline creating a REST api but that also means it does a lot for you in the background. These are things you can't easily change or write your own implementation for. For example, you might want to implement your own set of status codes, within the cRestDataset this is not currently possible.

If you wish to have a more custom implementation but still retain the possibility of generating the OpenApi specification you can use the cWebApiCustomEndpoint class. This class works similarly to the old cWebHttpHandler

(<https://docs.dataaccess.com/dataflexhelp/mergedprojects/VDFClassRef/cWebHttpHandler.htm>) .

Which means the developer is responsible for implementing the OnHttpGet, OnHttpPost, OnHttpPut, OnHttpPatch and OnHttpDelete events.

In order for the endpoint to be included in the OpenApi specification it is necessary to implement the OnDefineSchema event. This event comes with the endpointDefinition parameter. This struct needs to be filled in order for your endpoint to be reflected in the OpenApi specification.

For example, if you made a endpoint that exposes the GET and POST verbs, the GET verb retrieves the avatar of a person and the POST verb uploads a avatar of a person it could look something like this:

```
Procedure OnDefineSchema tEndpointDefinition ByRef
endpointDefinition
    tFieldDefinition imageField
    tVerbDefinition genericVerbInformation getVerb
postVerb
    tParameterDefinition personIDParam

//Configure the Image field
Move "image" to imageField.sFieldName
Move WEBAPI_BINARY_FIELD to imageField.eFieldType
Move True to imageField.bRequired

Move "ID" to personIDParam.sName
Move "ID of the person" to
personIDParam.sDescription
Move WEBAPI_QUERY_PARAMETER to
personIDParam.eParameterIn
Move WEBAPI_INTEGER_FIELD to
personIDParam.eParameterType
Move True to personIDParam.bRequired

Move personIDParam to
genericVerbInformation.parameters[-1]

//Configure the response data
Move imageField to
genericVerbInformation.responses[0].responseFields[-1]
Move C_WEBAPI_OK to
genericVerbInformation.responses[0].iStatusCode
Move "OK" to
genericVerbInformation.responses[0].sStatusCodeDescription
Move "application/octet-stream" to
genericVerbInformation.responses[0].asResponseMediaTypes[-
1]

//Move the shared info to the seperate structs
Move genericVerbInformation to getVerb
Move genericVerbInformation to postVerb

//Define the image field as a requested field
Move imageField to postVerb.requestFields[0]
```

The first step is to define a few structs. These do the following:

### **imageField**

This struct defines that image field that we expect from the client. This consists of a name, the type of the field and we can mark the field as required or not.

### **genericVerbInformation**

Inside of this struct we put the information that is shared between both the GET and POST verbs. That includes information such as the response format as they both return a avatar.

### **getVerb**

This struct contains the information needed to describe the GET verb.

### **postVerb**

This struct contains the information needed to describe the POST verb. This only slightly differs from the getVerb. The main differences are the description, the actual verb and the expected request body.

### **personIDParam**

This struct contains the information needed to describe the person id query parameter.

Next we start filling the structs with the needed information. The first thing we configure is the imageField struct. We give it the name "image" and mark it as a binary field. This is needed so the OpenApi specification understands it needs to use the file dialog. We also mark it as a required field.

Next we configure the person id parameter. First we set its name to "ID" and give it a short description. We mark it as a query parameter so that swagger knows this is a query parameter and not a path parameter. We mark the field type as integer and finally we mark it as a required field.

Next we configure the response format for both the verbs. We use the imageField struct we defined earlier as a response. You can define multiple responses for a single verb but for this example we'll stick to a singular response. If you wanted to create an additional response it is as simple as just using a different index in the genericVerbInformation.responses array. After we do this we add the status code 200 to the response. At last we mark "application/octet-stream" as the media type for the response.

Next we configure the differences between the GET and POST verbs. The main differences are in the actual verb and the description.

Aside from that the POST verb gets an additional expected request field from the client which is the image field. We also tell the POST verb that it can accept the "application/octet-stream" and "image/jpg" media types.

Finally we move the verbs we have defined into the endpointDefinition struct.

## 3 Components

This chapter explains per component what its functionality is inside of the system and in what way it communicates to other classes. Each subchapter will start with a short functional explanation of the class. Followed by a table with various properties and procedures that go into deeper technical detail about the class.

### 3.1 cWebApi

This class acts as the casing of the REST framework. This class extends from the cWebHttpHandler and is the class that will initially receive the http request. It forwards it by pulling apart the request path and forwarding it to a router with the path that is extracted from the URL.

The router will return a handle to the appropriate cRestDataset that should be used based on the current request. The cWebApi determines what iterator should be used for the current request, it passes this iterator to the cRestDataset class to build up a response. This class exposes a PreRequest and PostRequest event that developers can use to add functionality that should be fired before and after the event is handled.

If a cWebApiModifier is defined on this level it will be used on all incoming requests unless child objects opt out of inheriting the modifiers through the pbInheritSecurity property.

| cWebApi                   |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
| cWebHttpHandler           |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
| cRestModifierHost_Mixin   |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
| cWebApiRoutableHost_Mixin |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
| cWebApiErrorHandler_Mixin |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
| Property                  | Type                                                                                                                                                                                                                                                                                                | Description                                                                                                                                                                                                                |
| phoRoutables              | Handle[]                                                                                                                                                                                                                                                                                            | Handles to the registered routables.                                                                                                                                                                                       |
| pasRoutables              | String[]                                                                                                                                                                                                                                                                                            | The name of the registered routables, used to find the proper router to forward the message to.                                                                                                                            |
| piliteratorClasses        | Integer[]                                                                                                                                                                                                                                                                                           | References to the refclass of the iterator. Will later be used to create the iterator on runtime.                                                                                                                          |
| pasIteratorTypes          | String[]                                                                                                                                                                                                                                                                                            | The types of the different registered iterators. Will be used to determine what iterator should be used on the current request.                                                                                            |
| phoCachedIterator         | Handle                                                                                                                                                                                                                                                                                              | Handle to cache the last iterator used. If a subsequent request is send with the same content-type or accept-type this allows the iterator to be reused instead of creating a new one at runtime for every single request. |
| psApiTitle                | String                                                                                                                                                                                                                                                                                              | Property used to generate a title in the OpenApi specification.                                                                                                                                                            |
| psApiDescription          | String                                                                                                                                                                                                                                                                                              | Property used to generate a description in the OpenApi specification.                                                                                                                                                      |
| psApiVersion              | String                                                                                                                                                                                                                                                                                              | Property used to determine the version of the api. Will be used when generating the OpenApi specification.                                                                                                                 |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                         | Return type                                                                                                                                                                                                                |
| RegisterRoutables         | Registers a routable to the cWebApi. Routers need to be registered so that requests can be dynamically forwarded to them. This method is called inside of the cWebApiRouter or cRestDataset during the End_Construct_Object.<br><br><b>Params:</b><br>- Handle hoRoutable: The handle to the router |                                                                                                                                                                                                                            |

|                        |                                                                                                                                                                                                                                                                                                                                                                                   |  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| AddIterator            | <p>This procedure allows developers to register an iterator the the API.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Integer iRefClass: The refclass of the iterator. Will later be used to create an instance of the router at runtime.</li> <li>- String sMessageType: The message type of the iterator (like application/json).</li> </ul>              |  |
| OnHttpRequest          | Event that is called before the main functionality of the framework is performed.                                                                                                                                                                                                                                                                                                 |  |
| OnHttpPostRequest      | Event that is called after the main functionality of the framework is performed.                                                                                                                                                                                                                                                                                                  |  |
| GetIterator            | <p>Procedure that gets the appropriate iterator needed for a call. This is done by checking the accept type on GET and DELETE requests. For POST, PUT, and PATCH requests the content-type will be checked.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: The context to which the iterator should be added.</li> </ul> |  |
| CleanUp                | This procedure destroys old cached iterators if necessary.                                                                                                                                                                                                                                                                                                                        |  |
| FirePreModifierEvents  | <p>This procedure calls the OnPreRequest events from all the modifiers used in the current call.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>-tWebApiCallContext webapicalcontext: The context that holds the information of all the modifiers used in the current call.</li> </ul>                                                                          |  |
| FirePostModifierEvents | <p>This procedure calls the OnPostRequest events from all the modifiers used in the current call.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>-tWebApiCallContext webapicalcontext: The context that holds the information of all the modifiers used in the current call.</li> </ul>                                                                         |  |

## 3.2 cWebApiRouter

This class acts as a router within the framework. This class allows a developer to build in features such as versioning within their REST api. This could be done by having multiple cWebApiRouters within a cWebApi. Each router would have its own version like v1 and v2 and so on simply by setting a path property. The cWebApi can use this path and compare it to the path of the incoming request.

A cWebApiRouter can have nested routers. This allows for example a v1 router to have a public and private sub router. A developer can create as many nested routers as they want. The RouteRequest procedure in this class will recursively route requests to child routers until it no longer has a child router. This class forwards the request by looking at the path and then compares that to the path of its children. When it finds a matching one it forwards the request. This will be done until the request reaches a cRestDataset. At that point the router returns a handle of the cRestDataset to the cWebApi so it can start building up the response. If a cWebApiModifier is defined in this layer it will only be applicable for cRestDatasets defined within this cWebApiRouter and other child routers.

| cWebApiRouter             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| cObject                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| cRest_Mixin               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| cWebApiModifierHost_Mixin |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| cWebApiRoutableHost_Mixin |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Property                  | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Description |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Return type |
| RouteRequest              | <p>This procedure is called from the cWebApi. This procedure routes the request to the next object. The next object can either be a nested sub router or a cRestDataset. The request will keep being routed until it eventually reaches a cRestDataset. Each object that is used will be added to the webapicalcontext.</p> <p><b>Params:</b></p> <p>- tWebApiCallContext webapicalcontext: Struct that is passed through the pipeline ByRef with all the needed information.</p> | Handle      |

### 3.3 cBaseRestDataset

This class defines the data that will be exposed within a REST api. This class has all the base functionality needed for building a dataset.

| cBaseRestDataset          |                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
| cWebComponent             |                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
| cRest_Mixin               |                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
| cWebApiModifierHost_Mixin |                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
| Property                  | Type                                                                                                                                                                                                                                                                                                                                                | Description                                                                                   |
| pbReadOnly                | Boolean                                                                                                                                                                                                                                                                                                                                             | Determines if a dataset is read only and should only allow GET operations.                    |
| pbAllowRead               | Boolean                                                                                                                                                                                                                                                                                                                                             | Determines if a dataset allows GET requests.                                                  |
| pbAllowCreate             | Boolean                                                                                                                                                                                                                                                                                                                                             | Determines if a dataset allows POST requests.                                                 |
| pbAllowEdit               | Boolean                                                                                                                                                                                                                                                                                                                                             | Determines if a dataset allows PUT and PATCH requests.                                        |
| pbAllowDelete             | Boolean                                                                                                                                                                                                                                                                                                                                             | Determines if a dataset allows DELETE requests.                                               |
| pbSecureRead              | Boolean                                                                                                                                                                                                                                                                                                                                             | Property that determines if auth modifiers should act on GET requests to this endpoint.       |
| pbSecureCreate            | Boolean                                                                                                                                                                                                                                                                                                                                             | Property that determines if auth modifiers should act on POST requests to this endpoint.      |
| pbSecureEdit              | Boolean                                                                                                                                                                                                                                                                                                                                             | Property that determines if auth modifiers should act on PUT/PATCH requests to this endpoint. |
| pbSecureDelete            | Boolean                                                                                                                                                                                                                                                                                                                                             | Property that determines if auth modifiers should act on DELETE requests to this endpoint.    |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                                                                         | Return type                                                                                   |
| OnHttpGet                 | <p>This procedure is called when a GET request is send to the dataset. Should be augmented in subclasses</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.</li> </ul>                                                                      |                                                                                               |
| OnHttpGetById             | <p>This procedure is called when a GET request is send to the dataset. Should be augmented in subclasses</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.</li> <li>- Integer iID: The id of the source that needs to be found.</li> </ul> |                                                                                               |
| OnHttpPost                | <p>This procedure is called when a POST request is send to the dataset. Should be augmented in subclasses</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.</li> </ul>                                                                     |                                                                                               |
| OnHttpPut                 | <p>This procedure is called when a PUT request is send to the dataset. Should be augmented in subclasses</p>                                                                                                                                                                                                                                        |                                                                                               |

|                             |                                                                                                                                                                                                                                                                                                               |          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                             | <b>Params:</b><br>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.<br><br>- Integer iID: The id in the path parameter.                                                                                                                                               |          |
| OnHttpPatch                 | This procedure is called when a PATCH request is send to the dataset. Should be augmented in subclasses<br><br><b>Params:</b><br>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.<br><br>- Integer iID: The id in the path parameter.                                |          |
| OnHttpDelete                | This procedure is called when a DELETE request is send to the dataset. Should be augmented in subclasses<br><br><b>Params:</b><br>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.<br><br>- Integer iID: The id in the path parameter.                               |          |
| OnHttpRequest               | Entry point of this class. This delegates the request to the appropriate procedure. For example, a GET call will be called the OnHttpGet or OnHttpGetById procedure defined in this class.<br><br><b>Params:</b><br>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. |          |
| IsNumeric                   | Helper function to decide if a certain string is numeric.<br><br><b>Params:</b><br>-String sNumber: The string to check for a numeric value.                                                                                                                                                                  | Boolean  |
| GetDatasetTableName         | Helper function that retrieves the name of the current dataset.                                                                                                                                                                                                                                               | String   |
| RetrieveExposedDataFields   | Helper function that retrieves all the defined cRestField, cRestEntity and cRestChildCollection objects.                                                                                                                                                                                                      | Handle[] |
| GetFullEndpointPath         | Helper function that gets the full path of the endpoint. Up to the highest level cWebApiRouter object.                                                                                                                                                                                                        | String   |
| IsVerbAllowed               | Helper function that checks if the current verb is allowed based on the properties.<br><br><b>Params:</b><br>-String sVerb: The verb used in the current request.                                                                                                                                             | Boolean  |
| CurrentRecordToResponseBody | Gets the value of all the exposed data objects and uses the iterator to turn them into a response. For non data aware objects this calls the OnSetCalculatedValue.<br><br><b>Params:</b><br>-Handle hoResponseBody: Handle to the response body.<br>-Handle holterator: Handle to the iterator object.        |          |



### 3.4 cRestDataset

This class defines the data that will be exposed within a REST api. This class also calls other classes to build up a response to the client that initiated the request. This is done with the help of other classes such as the data dictionaries and iterator classes. To determine what fields will be exposed for the api consumers It uses a combination of the following classes:

- cRestField
- cRestEntity (For parent tables)
- cRestChildCollection (For child tables)

This class communicates with the data dictionary classes to find records, create records, alter records, and delete records. For example, during a GET request it will call the data dictionary class to load a record into the buffer. This class will then call procedures exposed by the cRestField, cRestEntity and cRestChildCollection classes to build up the response. Information that is retrieved by calling the cRestField, cRestEntity and cRestChildCollection classes is parsed into the response body using the iterator classes.

| cRestDataset         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                      |
| cBaseRestDataset     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                      |
| Property             | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Description                                                                                                                                                          |
| Main_DD              | Handle                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The handle to the main data dictionary used to perform find, save and delete operations.                                                                             |
| Server               | Handle                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The data dictionary to use as the server of this object.                                                                                                             |
| piLimitResults       | Integer                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Property that should be set after reading the url parameters. Determines how many results should be returned during a GET all request.                               |
| piFindIndex          | Integer                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | This property determines what index will be used for performing find operations on the data dictionary.                                                              |
| pbUsePathAsTableName | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | This property determines if the psPath should be used as table name in swagger. When this is set to false it will instead use the DF_FILE_LOGICAL_NAME of the table. |
| Procedure/Function   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Return type                                                                                                                                                          |
| OnHttpGet            | <p>This procedure returns all given records of the Main_DD specified in the object. It's possible to filter which data is retrieved by using query parameters in the request.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the hoResponseBody of this struct in combination with the holterator.</li> </ul>                             |                                                                                                                                                                      |
| OnHttpGetById        | <p>This procedure returns a single record of the Main_DD. The record that is to be retrieved is passed as an ID inside of the URL as a path parameter. Finds a record based on the exposed primary key field. If no primary key field is exposed this method will not work.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the</li> </ul> |                                                                                                                                                                      |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|               | <p>hoResponseBody of this struct in combination with the holterator.</p> <p>- Integer iID: The id of the source that needs to be found.</p>                                                                                                                                                                                                                                                                                                                                                                                       |  |
| OnHttpPost    | <p>This procedure is sent to save a record to the Main_DD. The response body is retrieved using either RequestDataString or RequestDataUChar.</p> <p><b>Params:</b></p> <p>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the hoResponseBody of this struct in combination with the holterator. The holterator is used to parse the request body to a tRESTRequestBody struct.</p>                                                                            |  |
| OnHttpPut     | <p>This procedure modifies a single record of the Main_DD in its entirety. The record that is to be changed is passed as an ID inside of the URL.</p> <p><b>Params:</b></p> <p>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the hoResponseBody of this struct in combination with the holterator. The holterator is used to parse the request body to a tRESTRequestBody struct.</p> <p>- Integer iID: The id of the resource that needs to be changed.</p> |  |
| OnHttpPatch   | <p>This procedure partially modifies a single record of the Main_DD specified by an ID inside of the URL.</p> <p><b>Params:</b></p> <p>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the hoResponseBody of this struct in combination with the holterator. The holterator is used to parse the request body to a tRESTRequestBody struct.</p> <p>- Integer iID: The id of the resource that needs to be changed.</p>                                         |  |
| OnHttpDelete  | <p>This procedure deletes a record from the Main_DD. The record to be deleted is specified by an ID in the URL.</p> <p><b>Params:</b></p> <p>- Integer iID: The ID of the resource that should be deleted.</p> <p>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the hoResponseBody of this struct in combination with the holterator.</p>                                                                                                                    |  |
| OnHttpRequest | <p>Entry point of this class. This delegates the request to the appropriate procedure. For example, a GET call will be called the OnHttpGet or OnHttpGetById procedure defined in this class.</p> <p><b>Params:</b></p>                                                                                                                                                                                                                                                                                                           |  |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                            | - tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. This procedure will fill the hoResponseBody of this struct in combination with the holterator.                                                                                                                                                                                                                                                                                                              |        |
| DataDictionaryErrorMessage | This function returns the last error thrown by dataflex.                                                                                                                                                                                                                                                                                                                                                                                                                                              | String |
| GetDatasetTableName        | Augmentation of the GetDatasetTableName from its super class. Checks the data dictionary for its table name.                                                                                                                                                                                                                                                                                                                                                                                          | String |
| RetrieveKeyField           | Helper function that retrieves the field that is linked to the tables primary key.                                                                                                                                                                                                                                                                                                                                                                                                                    | Handle |
| ResetState                 | This procedure clears the state of the main data dictionary.                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| HandleQueryParams          | <p>This procedure applies the constrains based on the query parameters sent in the current request. Is only used for GET requests.</p> <p>If you want to use constrains such as greater than, greater or equals you need to prefix the query param with for example (GT). So if you want to find all employees older than 24 your query parameter would look like the following:<br/>Age=(GT)24.</p> <p><b>Params:</b><br/>-Handle[] hoExposedDataObjects: All the exposed fields in the dataset.</p> |        |

### 3.5 cWebApiCustomEndpoint

This class allows developers to implement their custom logic inside of their REST api. The biggest difference this has compared to the regular cWebHttpHandler is that developers can define the information needed for the OpenApi specification by augmenting the OnDefineSchema procedure.

| cWebApiCustomEndpoint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Extends</b>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| cBaseRestDataset      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| Property              | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Description |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| Procedure/Function    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Return type |
| OnDefineSchema        | <p>This procedure allows a developer to define all the information needed for the OpenApi specification. This is done by filling the tEndpointDefinition struct with all the information needed.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tEndpointDefinition ByRef endpointDefinition: This struct should be filled with all the information needed for the endpoint. Configurations are on a verb basis. So a GET request could for example have completely different request fields and responses compared to a POST request. Security schemas are not needed in this struct. The framework will handle this.</li> </ul> | Handle      |

### 3.6 cWebApiLoginEndpoint

This class is used to allow developers a easy way to create login functionality for their REST apis. The default implementation uses the default session manager behaviour that is also used in web apps to facilitate logins. The login endpoint can be toggled between register and login mode with the pbRegisterMode property. When either a register or login is successful the developer can augment OnSuccessfulLogin to implement their custom logic. Much like the regular cRestDataset classes a developer can define fields that will be returned to the user. When a request is a success the OnSetCalculatedValue of these fields is called. Allowing developers to fill the field with the needed values.

For example, a developer could define a oSessionKey field. Whenever a login is a success inside of the OnSetCalculatedValue for this field the developer could move the WebAppSession.SessionKey to sValue. This will return the SessionKey to the user.

The cWebApiLoginEndpoint only exposes the POST verb.

| cWebApiLoginEndpoint |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Extends</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                           |
| cBaseRestDataset     |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                           |
| Property             | Type                                                                                                                                                                                                                                                                                                                                                                                                                                    | Description                                                               |
| pbRegisterMode       | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                 | Determines if the cWebApiLoginEndpoint is in register mode or login mode. |
| Procedure/Function   | Description                                                                                                                                                                                                                                                                                                                                                                                                                             | Return type                                                               |
| RegisterUser         | <p>This procedure attempts to register a new user through the web session manager. If the workspace does not have a session manager the developer will get a error, telling him/her to override the RegisterUser procedure with their own logic.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext ByRef webapicalcontext: This struct has all the information used in the current call.</li> </ul> | Boolean                                                                   |
| LoginUser            | <p>This procedure attempts to login a user through the web session manager. If the workspace does not have a session manager the developer will get a error, telling him/her to override the LoginUser procedure with their own logic.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext ByRef webapicalcontext: This struct has all the information used in the current call.</li> </ul>           | Boolean                                                                   |
| OnSuccessfulLogin    | <p>This event gets called whenever a login or register was a success. Developers can augment this event to implement some custom logic.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: This struct has all the information used in the current call.</li> </ul>                                                                                                                |                                                                           |

### 3.7 cOpenApiEndpoint

This is a endpoint that only exposes the OpenApi specification. The cSwaggerUI class makes a GET request to this endpoint to retrieve the OpenApi specification.

| cOpenApiEndpoint          |                    |                    |
|---------------------------|--------------------|--------------------|
| <b>Extends</b>            |                    |                    |
| cBaseRestDataset          |                    |                    |
| <b>Property</b>           | <b>Type</b>        | <b>Description</b> |
|                           |                    |                    |
| <b>Procedure/Function</b> | <b>Description</b> | <b>Return type</b> |
|                           |                    |                    |

### 3.8 cRestField

This class is nested inside of a cRestDataset and exposes a singular field. It is possible to define multiple cRestFields inside of a cRestDataset to expose multiple fields of a specific table.

It is not possible to retrieve all fields of a table with a singular cRestField. This is a design choice for security reasons. This prevents businesses from adding a field to a table that is then accidentally exposed through the REST api.

The field that is exposed is determined through an entry\_item like other data aware controls in DataFlex. Fields that should only be retrieved on GET requests but not be sent inside of POST requests can be marked as ReadOnly.

The main functionality of this class is to provide developers an easy way to expose table fields in their REST apis.

| cRestField          |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>      |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |
| cObject             |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |
| cBaseDEO_Mixin      |                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                      |
| Property            | Type                                                                                                                                                                                                                                                                                                                                                                                                                                    | Description                                                                                                                                                          |
| psFieldName         | String                                                                                                                                                                                                                                                                                                                                                                                                                                  | The name of the field is shown in the response body.                                                                                                                 |
| psExampleValue      | String                                                                                                                                                                                                                                                                                                                                                                                                                                  | This property determines what the example value is for the field. This will be shown in the OpenApi specification.                                                   |
| peFieldType         | Integer                                                                                                                                                                                                                                                                                                                                                                                                                                 | This property determines the field type of the current field.                                                                                                        |
| pbReadOnly          | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                 | Read only fields are only retrieved during GET requests and are omitted during other types of requests.                                                              |
| pbWriteOnly         | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                 | Write only fields are only used inside of POST, PUT and PATCH requests but are never returned in the response body.                                                  |
| pbFilterable        | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                 | This property determines if this field can be filtered through query parameters.                                                                                     |
| pbShowDuringGetAll  | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                 | This property determines if the field will be shown during a GET all. Allows developers to have less detail during a GET all compared to a GET of a specific record. |
| pbRequired          | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                 | This property determines if the field is required in the request body. This is used in the OpenApi specification.                                                    |
| Procedure/Function  | Description                                                                                                                                                                                                                                                                                                                                                                                                                             | Return type                                                                                                                                                          |
| AppendToBody        | <p>This procedure appends data to the response body. It does this in combination with the appropriate iterator.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle holterator: The iterator to use. The iterator is responsible for putting the value of the current field inside of a response body.</li> <li>- Handle hoResponseBody: The response body that is eventually sent back to the client.</li> </ul> |                                                                                                                                                                      |
| SetFieldGlobalValue | <p>This procedure changes the value of the field in the global buffers. Calls Set_Field_Value to change the value. This is mainly used to perform finds.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- String sValue: The value that the field should be changed to.</li> </ul>                                                                                                                                    |                                                                                                                                                                      |
| SetFieldLocalValue  | <p>This procedure changes the value of the field in the local buffers. Calls File_Field_Changed_Value to change the value. This way the field itself is responsible for changing its value and not the</p>                                                                                                                                                                                                                              |                                                                                                                                                                      |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|                      | <p>cRestDataset. This is used when doing POST, PUT and PATCH requests.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- String sValue: The value that the field should be changed to.</li> </ul>                                                                                                                                                                                                                              |              |
| AddConstrain         | <p>Adds a constrain to the current field if it is data aware.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- String sConstrain: The constrain to apply to the field.</li> <li>- String sFilterType: The filter type to use. Can be GE, GT, LT or LE. If nothing is specified will fall back to EQ.</li> </ul>                                                                                                               |              |
| IsKeyField           | <p>Checks if the current field is linked to the tables primary key. Returns true if this is the case otherwise returns false.</p>                                                                                                                                                                                                                                                                                                               | Boolean      |
| FieldValue           | <p>Retrieves the current value of the field.</p>                                                                                                                                                                                                                                                                                                                                                                                                | String       |
| FieldName            | <p>Retrieves the name of the field. If psFieldName is set this will be used and is absolute. If it is not set and the field is data aware it will retrieve the field name from the table.</p>                                                                                                                                                                                                                                                   | String       |
| FieldType            | <p>Retrieves the type of the field. For data aware fields this returns the field type from the table. For non data aware fields this returns peFieldType.</p>                                                                                                                                                                                                                                                                                   | Integer      |
| FieldHelp            | <p>Retrieves the help information for the current field. If psExampleValue is set this will be absolute. For data aware fields the File_Field_Status_Help is called from the data dictionary.</p>                                                                                                                                                                                                                                               | String       |
| FieldValidationTable | <p>Helper function that retrieves the validation table tied to the current field if there is one.</p>                                                                                                                                                                                                                                                                                                                                           | Variant[][2] |
| isFilterable         | <p>Returns if a field is filterable. This is done by retrieving pbFilterable and checking if it is false or true. If its set to true data aware fields will check if the current field is present in some kind of index.</p>                                                                                                                                                                                                                    | Boolean      |
| isRequired           | <p>Returns if a field is required. Data aware fields first check the required property of the data dictionary. If pbRequired is set to true this is absolute.</p>                                                                                                                                                                                                                                                                               | Boolean      |
| OnSetCalculatedValue | <p>This procedure is called on RestFields that do not have data binding. Gives the developer a opportunity to implement their own logic. This procedure is called right before the field is added to the response body.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- String ByRef sValue: sValue is the value of the field. Whatever the developer changes this to is what will be added to the response body.</li> </ul> |              |

### 3.9 cOpenApiRestField

Subclass of the cRestField. It's only purpose is returning the OpenApi specification inside of its OnSetCalculatedValue.

| cOpenApiRestField         |                    |                    |
|---------------------------|--------------------|--------------------|
| <b>Extends</b>            |                    |                    |
| cRestField                |                    |                    |
| <b>Property</b>           | <b>Type</b>        | <b>Description</b> |
|                           |                    |                    |
| <b>Procedure/Function</b> | <b>Description</b> | <b>Return type</b> |
|                           |                    |                    |

### 3.10 cRestChildCollection

This class allows for interaction with child tables. To use this class, you need to set the server of this object to that of the child data dictionary. It finds the child records based on the related field in the Main\_DD. It uses the same concept as the cWebList inside of DataFlex. You can define what fields of the child table in one of two ways.

The first option is to refer to a cRestDataset that exposes data of the child table. It will use the same data structure as defined in this endpoint.

The second option is nesting cRestFields inside of this object. The cRestFields determine what field of the child table will be exposed.

When the iterator sees this object, it knows how to structure it differently compared to regular cRestFields. The way it is structured is dependant on the iterator used.

Information provided in this class is only used during GET requests. The fields exposed in this class are omitted during POST, PUT, PATCH and DELETE requests. They are read only.

| cRestChildCollection |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                           |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Extends</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                           |
| cObject              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                           |
| cBaseDEO_Mixin       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                           |
| Property             | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Description                                                                               |
| psNodeName           | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The name of the nested array in the message body.                                         |
| pbReadOnly           | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Determines if this field is ReadOnly. Should always be true for child collections.        |
| piLimitResult        | Integer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Limits the amount of records returned by the cRestChildCollection. Set to 0 for no limit. |
| Procedure/Function   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Return type                                                                               |
| AppendToBody         | <p>This procedure appends data to the response body. It does this in combination with the appropriate iterator. For example, with a JSON iterator this procedure would create a nested JSON array inside of the current JSON object. It would then fill the JSON array by looping over the appropriate records and getting the field values from the nested cRestFields inside of this object.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoIterator: The iterator to use. The iterator is responsible for parsing the data and getting it to the response body.</li> <li>- Handle hoResponseBody: The response body that is eventually sent back to the client.</li> </ul> |                                                                                           |
| SchemaName           | Returns the name of the current cRestChildCollection. If psNodeName is set this is absolute. If this is not set the table name is returned for data aware cRestChildCollections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | String                                                                                    |
| FieldName            | Returns a empty string since cRestChildCollections use SchemaName instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | String                                                                                    |
| isFilterable         | Returns false since the framework does not support constrains on child tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Boolean                                                                                   |
| isRequired           | Returns false since the framework does not support posting of child records.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Boolean                                                                                   |

### 3.11 cRestEntity

This class allows for interaction with parent tables. It can find the data of the parent table related to this field. You can define the values retrieved from the parent table in two different ways.

The first option is to link to the cRestDataset that exposes the parent table. This way the structure that is defined there will be reused here.

The second option is to define cRestFields inside of this object with entry\_items that decide what field will be exposed.

When the iterator sees this object, it knows how to treat it differently than cRestFields and structure it differently. The way it is structured is dependant on the type of iterator.

These fields are only used when sending GET requests. The values exposed by this object are omitted during POST, PUT, PATCH and DELETE requests. They are read only.

This class behaves differently on POST, PUT and PATCH requests. During POST, PUT or PATCH requests it expects the foreign key used to link the two tables.

For example if we take the inventory table and the vendor table from the WebOrder database, If we have a cRestEntity object that represents the vendor table, whenever we make a POST request the body expects the Vendor\_ID field instead of the nested object entirely.

| cRestEntity        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Extends</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                    |
| cObject            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                    |
| cBaseDEO_Mixin     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                    |
| Property           | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Description                                        |
| psNodeName         | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | The name of the nested object in the message body. |
| pbReadOnly         | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Determines if this field is ReadOnly.              |
| Procedure/Function | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Return type                                        |
| AppendToBody       | <p>This procedure appends data to the response body. It does this in combination with the appropriate iterator. For example, with a JSON iterator this procedure would create a nested JSON object inside of the current JSON object. It would then fill the JSON object by looping over the appropriate records and getting the field values from the nested cRestFields inside of this object.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle holterator: The iterator to use. The iterator is responsible for parsing the data and getting it to the response body.</li> <li>- Handle hoResponseBody: The response body that is eventually sent back to the client.</li> </ul> |                                                    |
| SetFieldLocalValue | <p>Sets the value of the related field between the main data dictionary of the dataset and the server of this object.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- String sValue: The value to set.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                    |
| PrepareBinding     | <p>Simulates setting the entry_item command. This sets the data_file and data_field to the field of the main data dictionary that relates to the server set to this object.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                    |

|                            |                                                                                                                                |         |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------|
| FieldName                  | Returns the name of the field that manages the connection between the main table and the server of this cRestEntity.           | String  |
| SchemaName                 | Returns the name of this entity. Either returns psNodeName if it is set. Otherwise will retrieve the table name of the server. | String  |
| IsFilterable               | Returns false since the framework does not support constraints on parent tables.                                               | Boolean |
| isRequired                 | Returns false since the framework does not support required fields on parent tables.                                           | Boolean |
| RetrieveServerRelatedField | Helper function that retrieves the field from the main table that is linked to this cRestEntity's server.                      | Integer |

### 3.12 cWebApiModifier

This class allows a developer to add different types of modifiers to their program. A modifier can be a lot of different things, but it is designed with the idea that it is used for authentication/authorization and logging. It adds hooks that developers themselves can extend upon and add their own functionality. As the name suggests this is meant to be the most modifyable part of the framework.

If a developer wishes to implement security mechanisms into their API they should look into implementing the cWebApiAuthModifier class instead as it has more logic to implement security mechanisms more easily.

| cWebApiModifier    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Extends</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| cObject            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| Property           | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| Procedure/Function | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Return type |
| OnPreRequest       | <p>The event that is triggered before handling the HTTP request. This event would be the place to build in mechanisms such as authorization and authentication. It would also be an appropriate place to build in mechanisms such as logging when a request comes in.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef.</li> </ul>                                                                                                         |             |
| OnPostRequest      | <p>This event is triggered when the framework is done handling the main logic for the HTTP request. This would be an appropriate place to build in mechanisms such as logging how the message was handled. Things such as status code, time spent processing and more.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. Can be useful for logging modifiers to retrieve the status codes and other information of the request.</li> </ul> |             |

### 3.13 cWebApiAuthModifier

A subclass of the cWebApiModifier that implements extra functionality to allow implementing authentication or authorization needs of developers more easily. During object creation this class builds up a list of cRestDatasets that it secures. This is done by parsing objects that are defined on the same level as this object or objects that are defined on a lower level. Objects can opt out of using their parents' security mechanisms by using the property pblInheritSecurity. This class modifies the OnPreRequest behaviour. Whenever the OnPreRequest of this object is called it will check if the endpoint used in the current call is present in this objects list of secured endpoints. If it is the OnAuth event will be called.

| cWebApiAuthModifier  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                         |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                         |
| cWebApiModifier      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                         |
| Property             | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description                                                                                                                                                                             |
| psSecuritySchemaName | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Property used during generation of the OpenApi specification. This is a mandatory property if the developer opts into generating the OpenApi specification.                             |
| pvSecurityInfo       | Variant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | This should be filled with a struct that matches the security schema of the used authentication scheme. These schemas can be found in the OpenApi documentation on the swagger website. |
| ptSecuredDatasets    | tSecuredDataset[]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Property with a list of secured endpoints. Will be used during the PreRequest to determine if the OnAuth request should be triggered.                                                   |
| Procedure/Function   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Return type                                                                                                                                                                             |
| OnAuth               | <p>This event can be triggered during the PreRequest event. The cWebApiAuthModifier will compare the handle of the cRestDataset in the current request to the handles that are inside of ptSecuredDatasets.</p> <p>Developers should implement their security logic inside of this event and set the variable webapicalcontext.bErr to either true if the consumer does not have access to the current resource or False if they do have access to this resource.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: Struct that is passed through the framework ByRef. Developers can augment the bErr, iStatusCode, sShortStatusMessage and sErrorMessage of this class to block access to a certain resource.</li> </ul> |                                                                                                                                                                                         |
| InitSecuredEndpoints | This procedure is called in the End_Construct_Object of its parent class. It walks through the structure of the framework and adds all datasets that use the security mechanism of this class to a list. This list will later be used during requests to determine if the OnAuth event needs to be called.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                         |
| RecursiveSecure      | <p>This procedure recursively secures child objects. Keeps going down up until it finds a dataset to secure.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoParent: The last visited object in this recursive call.</li> <li>- tSecuredDataset[] ByRef securedDatasets: A list of all the secured datasets that is passed byref.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                         |
| DefineAuthRules      | This procedure calls the OnDefineAuthRules event and allows the developer to define their own authentication schema. This                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                         |

|                   |                                                                                                                                                                                                                   |  |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                   | eventually sets the pvSecurityInfo property to the struct defined by the developer. Developers should augment this event if they intend to have their cWebApiAuthModifier reflected in the OpenApi specification. |  |
| OnDefineAuthRules | Developers should augment this event and move their own struct to the vSecurityStruct variable. The framework uses this variable to generate the appropriate OpenApi specification.                               |  |

### 3.14 cBaseWebApiliterator

This is the base class for iterators. An iterator is an object that is used to formulate a response body in a specific type. It can also be used to convert a request body to a tRESTRestBody struct that is easier to use in DataFlex code. All iterators should inherit from

This class covers the shared functionality for each iterator. It's meant as a base class that can be inherited from and built upon. Classes that extend this class should override the functionality of this class and implement it in their own datatypes.

| cBaseWebApiliterator      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                     |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                     |
| cObject                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                     |
| <b>Property</b>           | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Description</b>                                                                                                                                                                                                                                                                                  |
| psMessageType             | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | This defines the type of message the iterator covers. Sub classes would set it to specify what data type they are used for. For example, this could be set to JSON to specify that the iterator is used to build up and parse JSON objects. This should match the value in the Content-Type header. |
| <b>Procedure/Function</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Return type</b>                                                                                                                                                                                                                                                                                  |
| ModifyResponseBody        | <p>This procedure is used to build up the response body back to the client. The values needed for this procedure to work are retrieved from the cRestField, cRestChildCollection and cRestEntity if they are declared.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoResponseBody: The response body that is eventually sent back to the client. The values are appended to this body.</li> <li>- Variant vValue: The value that should be appended to the responsebody. Because this value could be of any type variant is used.</li> <li>- String sDataType: The datatype that vValue has. This is used to sow the correct data type in the response body. For example, in a JSON response passing "integer" as a datatype causes jsonTypeInteger to be used when appending members.</li> <li>- String sFieldName: The name of the current field.</li> </ul> |                                                                                                                                                                                                                                                                                                     |
| ParseRequestBody          | <p>This function is used to parse the request body into a data type that is easily understandable by DataFlex as a key-value pair. Because DataFlex does not have hashmaps a struct of type tRESTRestBodyStruct is returned.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoRequestBody: The request body that is send from the client. This will be parsed into a tRESTRestBodyStruct[[]].</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | tRESTRestBodyStruct[[]]                                                                                                                                                                                                                                                                             |
| PrepareForTransfer        | This procedure prepares the response body to be returned to the client. This should be augmented in sub classes to stringify their structures so that it can be sent over http.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                     |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                          | <b>Params:</b><br>- tWebApiCallContext webapicalcontext: Provides the iterator with all information. This includes a handle to the response body, status codes and potential error messages                                                                                                                                                                                                                                                                                                                        |        |
| CreateResponseBodyArray  | This function creates an array of a certain datatype. Should be augmented inside of the sub classes.<br><br><b>Params:</b><br>- String sTableName: The table name, will be used to name the array.                                                                                                                                                                                                                                                                                                                 | Handle |
| CreateResponseBodyObject | This function creates an object of a certain datatype. Should be augmented inside of sub classes.<br><br><b>Params:</b><br>- String sTableName: The table name, will be used to name the object.                                                                                                                                                                                                                                                                                                                   | Handle |
| AppendToResponseArray    | This procedure allows an object to be appended to an array of a certain datatype. Should be augmented inside of sub classes.<br><br><b>Params:</b><br>- Handle hoNestedObject: The object that should be appended to the array.<br><br>- Handle hoResponseArray: The array that the object should be appended to.                                                                                                                                                                                                  |        |
| GenerateErrorResponse    | This procedure formulates an error response back to the client. Uses the status code and error message to parse into a response object. Errors are formatted according to the following RFC: <a href="https://www.rfc-editor.org/rfc/rfc9457.html">https://www.rfc-editor.org/rfc/rfc9457.html</a><br><br><b>Params:</b><br>- tWebApiCallContext webapicalcontext: Provides this method all information related to the request. Including a handle to the response body, status codes and potential error messages |        |
| SetContentType           | This procedure sets the content-type for the http request that will be send back to the client. Uses the psMessageType property to determine what to set in the header.                                                                                                                                                                                                                                                                                                                                            |        |

### 3.15 cJSONIterator

This class is used to retrieve and build data into a JSON format. It works in collaboration with cRestFields, cRestChildCollections and cRestEntity's to build up a response back to the client. It can also be used to parse a JSON request body to a datatype that is easily useable in DataFlex code. This is the tRESTRequestBody struct.

When it runs into a cRestField it will create a regular JSON member. When it runs into a cRestChildCollection it will create a JSON array. When it runs into a cRestEntity it will create a nested JSON object.

| cJSONIterator             |                                                                                                                                                                                                                                                                                                                                                                  |             |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                                                                                  |             |
| cBaseWebApiliterator      |                                                                                                                                                                                                                                                                                                                                                                  |             |
| Property                  | Type                                                                                                                                                                                                                                                                                                                                                             | Description |
|                           |                                                                                                                                                                                                                                                                                                                                                                  |             |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                                                                                      | Return type |
| RecursiveJsonBodyToStruct | <p>This procedure recursively parses the request body into a tRESTRequestBody struct array.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoJson: The current JSON object to parse.</li> <li>- tRESTRequestBody[] ByRef requestBody: the struct that will be filled with all the information related to the request body.</li> </ul> |             |

### 3.16 cXMLIterator

This class is used to build and retrieve data into an XML format. It works in collaboration with the cRestFields, cRestChildCollections and cRestEntity's to build up a response back to the client. Apart from building up a response to the client it is also capable of parsing the XML request body from the client to an easy to use datatype in DataFlex. This is the tRESTRequestBody struct.

When it runs into a cRestField it will create an XML node. When it runs into a cRestEntity it will create a nested XML node. When it runs into a cRestChildCollection it will create a nested XML node with multiple child nodes.

| cXMLIterator              |                                                                                                                                                                                                                                                                                                                                               |                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                                                               |                                                               |
| cBaseWebApiliterator      |                                                                                                                                                                                                                                                                                                                                               |                                                               |
| <b>Property</b>           | <b>Type</b>                                                                                                                                                                                                                                                                                                                                   | <b>Description</b>                                            |
| phoXmlDocument            | Handle                                                                                                                                                                                                                                                                                                                                        | The outer most xml object. Used to create nested xml objects. |
| <b>Procedure/Function</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                            | <b>Return type</b>                                            |
| RecursiveParseXML         | <p>Recursively parses the xml request body into a struct.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>-Handle hoNode: The current xml node that needs to be parsed.</li> <li>- tRESTRequestBody[] ByRef requestbody: The struct that needs to be filled with all the information related to the request body.</li> </ul> |                                                               |

### 3.17 cRest\_Mixin

This mixin covers the functionality shared between the cWebApiRouter and cRestDataset. This covers the psPath property and the Locate\_Server and Server functions. These functions have to be implemented into these objects to allow the framework to operate within a cWebAppBasic project.

| cRest_Mixin        |                                                                                        |                              |
|--------------------|----------------------------------------------------------------------------------------|------------------------------|
| <b>Extends</b>     |                                                                                        |                              |
| Mixin              |                                                                                        |                              |
| Property           | Type                                                                                   | Description                  |
| psPath             | String                                                                                 | Part of the path in the URL. |
| Procedure/Function | Description                                                                            | Return type                  |
| Locate_Server      | Function that is included to allow the framework to work within cWebAppBasic projects. | Handle                       |
| Server             | Function that is included to allow the framework to work within cWebAppBasic projects. | Handle                       |

### 3.18 cWebApiModifierHost\_Mixin

This mixin implements the functionality needed to attach modifiers to a object.

| cWebApiModifierHost_Mixin |                                                                                                                                                                                                                                                                                                             |                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                             |                                                                                     |
| Mixin                     |                                                                                                                                                                                                                                                                                                             |                                                                                     |
| Property                  | Type                                                                                                                                                                                                                                                                                                        | Description                                                                         |
| phoModifiers              | Handle[]                                                                                                                                                                                                                                                                                                    | A list of the registered modifiers.                                                 |
| pblInheritSecurity        | Boolean                                                                                                                                                                                                                                                                                                     | Property that determines if a object will inherit the auth modifiers of its parent. |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                                 | Return type                                                                         |
| RegisterModifier          | <p>This procedure allows a modifier to be registered to the current object. This is called from the cWebApiModifier during its End_Construct_Object.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoModifier: The handle to the modifier that should be registered.</li> </ul> |                                                                                     |
| AddModifiersToCallContext | <p>This procedure allows a object that implements this mixin to easily add its current modifiers to the webapicalcontext.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- tWebApiCallContext webapicalcontext: The struct that has all information about the current request.</li> </ul> |                                                                                     |
| GetAuthModifiers          | This helper function retrieves all the authentication modifier for the current object.                                                                                                                                                                                                                      | Handle[]                                                                            |

### 3.19 cWebApiRoutableHost\_Mixin

This mixin has all the functionality needed for the routing logic of the framework. Objects that implement this mixin have the ability to register child routables. The current routable hosts in the framework are the cWebApi and the cWebApiRouter.

| cWebApiRoutableHost_Mixin |                                                                                                                                                                                                                                                                                                              |                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                              |                                                                                                                |
| Mixin                     |                                                                                                                                                                                                                                                                                                              |                                                                                                                |
| Property                  | Type                                                                                                                                                                                                                                                                                                         | Description                                                                                                    |
| phoRoutables              | Handle[]                                                                                                                                                                                                                                                                                                     | Handle to all the child routables.                                                                             |
| pasRoutables              | String[]                                                                                                                                                                                                                                                                                                     | Names of all the child routables. When searching for a routable we first search the string array if it exists. |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                                  | Return type                                                                                                    |
| RegisterRoutable          | <p>This procedure is called by a child cWebApiRouter or cBaseRestDataset. It registers the routable to the current object. Allowing for message routing.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoRoutable: Handle to the child routable object.</li> </ul>               |                                                                                                                |
| RoutableIndex             | <p>This procedure binary searches the pasRoutables array to see if the current object has a routable that matches the sPath. If found returns the index. If nothing is found it will return -1.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- String sPath: The request URL.</li> </ul> | Integer                                                                                                        |

### 3.20 cWebApiErrorHandler\_Mixin

This mixin is used to handle unexpected errors that might occur during a http call to the service. Incase one of these unexpected errors might occur the server will return a generic error telling the client "Something went wrong on the server". If you want more details on what went wrong you can toggle pbDebugMode to true. This will instead return the callstack when a unexpected error occurs.

| cWebApiErrorHandler_Mixin |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                               |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>            |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                               |
| Mixin                     |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                               |
| Property                  | Type                                                                                                                                                                                                                                                                                                                                                                                         | Description                                                                                                                                                                                                                   |
| pbUnexpectedError         | Boolean                                                                                                                                                                                                                                                                                                                                                                                      | A flag used to remember if a unexpected error occurred during the request.                                                                                                                                                    |
| pasErrorCallstack         | String[]                                                                                                                                                                                                                                                                                                                                                                                     | A array of callstacks, everytime a error occurs during the current request the callstack is pushed to this array when pbDebugMode is set to true.                                                                             |
| pbDebugMode               | Boolean                                                                                                                                                                                                                                                                                                                                                                                      | Determines how error messages are returned to the client. When running in pbDebugMode the entire callstack is returned to the client. When pbDebugMode is false the message "Something went wrong on the server" is returned. |
| Procedure/Function        | Description                                                                                                                                                                                                                                                                                                                                                                                  | Return type                                                                                                                                                                                                                   |
| Error_Report              | <p>This procedure is called by the error handler whenever a error occurs. Writes the callstack to pasErrorCallStack when pbDebugMode is true.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Integer ErrNum: The number of the error.</li> <li>- Integer Err_Line: The line on which the error occurred.</li> <li>- String sErrMsg: The actual error message.</li> </ul> |                                                                                                                                                                                                                               |
| StartErrorTracking        | This procedure allows the cWebApi object to register itself as the error object during the current request.                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                               |
| StopErrorTracking         | This procedure makes the ghoErrorHandler the error object again and resets all variables to their default values.                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                               |
| HttpErrorMessage          | This function gets the correct error message based on what mode the error handler is running in. In pbDebugMode this will return the entire callstack. When not running in pbDebugMode this returns "Something went wrong on the server"                                                                                                                                                     | String                                                                                                                                                                                                                        |
| DetailedErrorMessage      | Helper function that combines pasErrorCallstack into a singular string that can be returned to the client.                                                                                                                                                                                                                                                                                   | String                                                                                                                                                                                                                        |

### 3.21 cOpenApiSpecification

This class is responsible for building the OpenApi specification in JSON format. This is done by iterating through the existing structure and parsing the info of the objects into a JSON file using Direct\_Output. Each object in the framework maps to a different part of the OpenApi specification. This class is created after the cWebApi finishes initializing. It's main procedure will be called and after it is done generating the OpenApi specification it will be destroyed again.

This class is considered private.

| cOpenApiSpecification        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Extends</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| cObject                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| Property                     | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description |
|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| Procedure/Function           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Return type |
| GenerateOpenApiSpecification | This is the main procedure of this class and will start creating the json object that holds all the info about the OpenApi specification. This will call the other procedures in this class to formulate the OpenApi specification.                                                                                                                                                                                                                                              | String      |
| GenerateServerInfo           | This procedure generates all the info that should be present inside of the "Servers" JSON array in the OpenApi specification.<br><br><b>Params:</b><br>- Handle hoOpenApiSpecJson: A handle to the OpenApi specification JSON.                                                                                                                                                                                                                                                   |             |
| GenerateEndpointInfo         | This procedure generates all the info about a endpoint. This should generate both the information inside of the "Paths" JSON object in the OpenApi specification as well as the "schemas" JSON object defined inside of the "components" JSON object.<br><br><b>Params:</b><br>- Handle hoOpenApiSpecJson: A handle to the OpenApi specification JSON.                                                                                                                           |             |
| GenerateErrorSchema          | This procedure generates the structure for the error responses inside of the "schemas" JSON object of the OpenApi specification. This should only be done once because the framework reuses the error template everywhere.<br><br><b>Params:</b><br>- Handle hoSchemasJson: A handle to the schema object inside of the OpenApi specification.                                                                                                                                   |             |
| GenerateSecurityInfo         | This procedure generates the security objects inside of the OpenApi specification. It does this by placing information of the security modifier inside of the "securitySchemes" JSON object of the OpenApi specification.<br><br>After the information inside of the "securitySchemes" JSON has been filled it will find the JSON objects of the applicable endpoints that use the security mechanism. It will insert information about itself into the JSON of these endpoints. |             |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                     | <b>Params:</b><br>-Handle hoOpenApiSpec: Handle to the OpenApi specification object. This is needed to get all the endpoint information.                                                                                                                                                                                                                                                                                                                                                         |  |
| ParseEndpoint       | This procedure parses the information of a endpoint into the OpenApi specification.<br><br><b>Params:</b><br>- Handle hoPathsJson: A handle to the paths object inside of the OpenApi specification.<br>- Handle hoEndpoint: Handle to the endpoint to be parsed.                                                                                                                                                                                                                                |  |
| ParseVerb           | This procedure parses a verb (GET, POST, PUT, PATCH AND DELETE) from a specific endpoint.<br><br><b>Params:</b><br>- String sCurrentVerb: The current verb to parse.<br>- Handle hoEndpointJson: Handle to the json object that has all the information related to the current endpoint. The verb information will be appended to this.<br>- Handle hoEndpoint: Handle to the endpoint to be parsed.                                                                                             |  |
| ParseSchema         | This procedure parses the data structure of a endpoint into the schema section of the OpenApi specification.<br><br><b>Params:</b><br>- Handle hoSchemasJson: Handle to the schema section object of the OpenApi specification.<br>- Handle hoEndpoint: Handle to the endpoint to be parsed.                                                                                                                                                                                                     |  |
| ParseCustomEndpoint | This procedure parses the endpoint that are of the class "cWebApiCustomEndpoint".<br><br><b>Params:</b><br>- Handle hoPathsJson: Handle to the paths object of the OpenApi specification.<br>- Handle hoEndpoint: Handle to the endpoint to be parsed.                                                                                                                                                                                                                                           |  |
| ParseCustomVerb     | This procedure parses a verb from a custom endpoint. It does this by reading the tVerbDefinition struct.<br><br><b>Params:</b><br>- Handle hoEndpointJson: Handle to the endpoint object inside of the OpenApi specification. Verb information will be appended to this.<br>- String sPath: The path of the current endpoint.<br>- tVerbDefinition verbDefinition: Holds all the information related to the current verb. It is up to developers to fill this struct with the right information. |  |
| ParseCustomField    | This procedure parses a custom defined field.<br><br><b>Params:</b><br>- tFieldDefinition currentField: The information of the current field. It is up to developers to implement this and add the right information.<br>- Handle hoPropertiesJson: Handle to the properties part of the OpenApi specification.                                                                                                                                                                                  |  |
| ParseErrorResponse  | This procedure parses the generic error response per verb.                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                           | <b>Params:</b> <ul style="list-style-type: none"> <li>- String sVerb: The verb that needs to be parsed.</li> <li>- Handle hoResponsesJson: Handle to the responses object for the current verb.</li> </ul>                                                                                                                                                                                                                                                                            |  |
| ApplyQueryParams          | <p>This procedure adds the available query parameters to the GET verb.</p> <b>Params:</b> <ul style="list-style-type: none"> <li>- Handle hoVerbJson: Handle to the verb information JSON object.</li> <li>- Handle hoEndpoint: Handle to the current endpoint.</li> </ul>                                                                                                                                                                                                            |  |
| FieldToOpenApi            | <p>This procedure parses a field that is defined inside of a cRestDataset to the OpenApi specification.</p> <b>Params:</b> <ul style="list-style-type: none"> <li>- Handle hoField: Handle to the current field that needs to be parsed into the OpenApi specification.</li> <li>- Handle hoPropertiesJson: Handle to the properties JSON object that this field needs to be appended to.</li> </ul>                                                                                  |  |
| AddValidationTableToField | <p>This procedure checks if a field has a validation table. If it does the necessary information will be added to the OpenApi specification.</p> <b>Params:</b> <ul style="list-style-type: none"> <li>- Handle hoFieldJson: Handle to the JSON object that has all the information of the current field.</li> <li>- Variant[][] avValidationTable: The information of the validation table.</li> </ul>                                                                               |  |
| RecursiveParseDatasets    | <p>This procedure goes over all the endpoints inside of the api and calls ParseSchema, ParseVerb and other procedures where necessary.</p> <b>Params:</b> <ul style="list-style-type: none"> <li>- Handle hoPathsJson: Handle to the JSON object that holds all the information related to paths.</li> <li>- Handle hoSchemasJson: handle to the JSON object that holds all the information related to schemas.</li> <li>- Handle hoRoutable: The current routable object.</li> </ul> |  |
| GetAllAuthModifiers       | <p>Helper function that gets all the cWebApiAuthModifiers from every routable object. Whenever the object is not a cBaseRestDataset the procedure recursively calls this procedure on all child routables too.</p> <b>Params:</b> <ul style="list-style-type: none"> <li>- Handle hoCurrentObject: The current object that has to be checked.</li> <li>- Handle[] ByRef hoAuthModifiers: Handle array that holds all the cWebApiAuthModifiers.</li> </ul>                             |  |
| FieldTypeToOpenApiType    | <p>Helper function that converts a dataflex type to the correct type for the OpenApi specification.</p> <b>Params:</b> <ul style="list-style-type: none"> <li>- Integer eDataflexFieldType: The Dataflex field type.</li> </ul>                                                                                                                                                                                                                                                       |  |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|              | <ul style="list-style-type: none"> <li>- String ByRef sFieldType: The type of the field needed for the OpenApi specification</li> <li>- String ByRef sFieldFormat: The formatting of the field. Binary fields for example require binary formatting in the OpenApi specification.</li> </ul>                                                                                                                           |        |
| EndpointPath | <p>Helper function that generates a path for the current endpoint. Calls GetFullEndpointPath from the current endpoint and then checks if a Id needs to be appended to the URL.</p> <p><b>Params:</b></p> <ul style="list-style-type: none"> <li>- Handle hoEndpoint: The endpoint of which the path should be retrieved.</li> <li>- Boolean bIdRequired: Determines if a Id should be appended to the URL.</li> </ul> | String |

### 3.22 cSwaggerUI

This class is responsible for rendering the OpenApi specification on the client. Developers can drag this custom control in any view they desire. This control renders the OpenApi specification by sending a http request to psOpenApiUrl.

| cSwaggerUI                |                    |                                                                                                                               |
|---------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Extends</b>            |                    |                                                                                                                               |
| cWebBaseControl           |                    |                                                                                                                               |
| <b>Property</b>           | <b>Type</b>        | <b>Description</b>                                                                                                            |
| psOpenApiUrl              | String             | The client sends a request to this url to retrieve the OpenApi specification. This Url is relative from your application Url. |
| <b>Procedure/Function</b> | <b>Description</b> | <b>Return type</b>                                                                                                            |
|                           |                    |                                                                                                                               |

## 4 Structs, constants and enums

### 4.1 Structs

#### 4.1.1 tRESTRestRequestBody

This struct is used when parsing data from a request body to a useable type. For example when someone does a post request with a JSON body the cJSONIterator will parse the body into a tRESTRestRequestBody array. This struct can then be used to save the data to the proper fields. This struct is used as a substitution for a hash map because those do not exist in DataFlex as of the moment of creating this framework.

| tRESTRestRequestBody |                        |                                                                                       |
|----------------------|------------------------|---------------------------------------------------------------------------------------|
| Member name          | Type                   | Description                                                                           |
| sFieldName           | String                 | The name of the field.                                                                |
| sFieldValue          | String                 | The actual value of the field.                                                        |
| nestedFields         | tRESTRestRequestBody[] | Potentially nested fields. (only happens in for example a json object or json array.) |

#### 4.1.2 tWebApiCallContext

This struct contains all the information of the current http request. This struct is passed through the request pipeline ByRef and is filled as it passes through each object. This file also defines constants used throughout the framework.

| tWebApiCallContext  |          |                                                                                                                                                                                                                                |
|---------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member name         | Type     | Description                                                                                                                                                                                                                    |
| hoApi               | Handle   | Handle to the cWebApi object.                                                                                                                                                                                                  |
| hoDataset           | Handle   | Handle to the cRestDataset object used inside of the call.                                                                                                                                                                     |
| hoRouters           | Handle[] | Handle to all the routers used inside of the request.                                                                                                                                                                          |
| hoResponseBody      | Handle   | Handle to the response body of the current request.                                                                                                                                                                            |
| holterator          | Handle   | Handle to the iterator used during the current request.                                                                                                                                                                        |
| hoModifiers         | Handle[] | Handle to the iterator used inside of the current request.                                                                                                                                                                     |
| bErr                | Boolean  | Boolean that is set to true if something has gone wrong during the request pipeline. The actual error sent back to the client is made up of a combination of the iStatusCode, sShortStatusMessage and sErrorMessage variables. |
| bIsCustom           | Boolean  | Boolean that is set to true if the request is send towards a custom endpoint. If this is set to true the iterator classes will not be used to formulate a response back to the client.                                         |
| iStatusCode         | Integer  | Status code of the current request.                                                                                                                                                                                            |
| sShortStatusMessage | String   | Short status message to compliment the status code. For example "OK" or "Forbidden"                                                                                                                                            |
| sErrorMessage       | String   | A longer error message displayed in the error response that the client receives.                                                                                                                                               |
| sPath               | String   | The request path of the current request.                                                                                                                                                                                       |
| sVerb               | String   | The verb of the current request.                                                                                                                                                                                               |
| sContentType        | String   | The content-type of the current request if applicable.                                                                                                                                                                         |
| sAcceptType         | String   | The accept-type of the current request if applicable.                                                                                                                                                                          |
| sMainTableName      | String   | The name of the main table used in the endpoint that is handling the current request.                                                                                                                                          |

#### 4.1.3 tSecuredDataset

The cWebApiAuthModifier maintains a list of this struct to determine what endpoints are secured and what verbs of that specific endpoint are secured.

| tSecuredDataset |         |                                                                  |
|-----------------|---------|------------------------------------------------------------------|
| Member name     | Type    | Description                                                      |
| hoDataset       | Handle  | Handle to the dataset.                                           |
| bSecureRead     | Boolean | Determines if the GET verbs of a endpoint are secured.           |
| bSecureCreate   | Boolean | Determines if the POST verbs of a endpoint are secured.          |
| bSecureEdit     | Boolean | Determines if the PUT and PATCH verbs of a endpoint are secured. |
| bSecureDelete   | Boolean | Determines if the DELETE verbs of a endpoint are secured.        |

#### 4.1.4 oneOf

This struct is used to parse validation tables into the OpenApi specification.

| tSecuredDataset |        |                                                |
|-----------------|--------|------------------------------------------------|
| Member name     | Type   | Description                                    |
| title           | String | The long value of the validation table entry.  |
| const           | String | The short value of the validation table entry. |
| description     | String | The long value of the validation table entry.  |

#### 4.1.5 tEndpointDefinition

This struct holds multiple tVerbDefinitions. A developer can fill the value of this struct to determine what will be shown in the OpenApi specification.

| tEndpointDefinition |                   |                                                          |
|---------------------|-------------------|----------------------------------------------------------|
| Member name         | Type              | Description                                              |
| verbDefinitions     | tVerbDefinition[] | Information about all the verbs in the current endpoint. |

#### 4.1.6 tVerbDefinition

This struct holds all the information relevant to a specific verb in a endpoint.

| tVerbDefinition     |                        |                                                                      |
|---------------------|------------------------|----------------------------------------------------------------------|
| Member name         | Type                   | Description                                                          |
| sVerb               | String                 | The name of the verb.                                                |
| SDescription        | String                 | The description of the verb to be used in the OpenApi specification. |
| Responses           | tResponseDefinition[]  | All the potential responses for the current verb.                    |
| asRequestMediaTypes | String[]               | The supported media types for requests.                              |
| requestFields       | tFieldDefinition[]     | Information about all the exposed request fields.                    |
| Parameters          | tParameterDefinition[] | All parameters used for the verb. Can be query or header parameters. |

#### 4.1.7 tResponseDefinition

This struct has all the information related to a response.

| tResponseDefinition    |                    |                                                             |
|------------------------|--------------------|-------------------------------------------------------------|
| Member name            | Type               | Description                                                 |
| iStatusCode            | Integer            | The status code that is linked to this response.            |
| sStatusCodeDescription | String             | The description of the status code.                         |
| asResponseMediaTypes   | String[]           | The response types available.                               |
| responseFields         | tFieldDefinition[] | The fields that are returned to the consumer in a response. |

#### 4.1.8 tFieldDefinition

This struct has all the information related to defining a field in the OpenApi specification.

| tFieldDefinition |                    |                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member name      | Type               | Description                                                                                                                                                                                                                                                                                                                                                                        |
| sFieldName       | String             | Name of the field.                                                                                                                                                                                                                                                                                                                                                                 |
| sExampleValue    | String             | Example value shown in the OpenApi specification.                                                                                                                                                                                                                                                                                                                                  |
| eFieldType       | Integer            | Type of the field can be one of the following:<br><ul style="list-style-type: none"> <li>- WEBAPI_INTEGER_FIELD</li> <li>- WEBAPI_NUMBER_FIELD</li> <li>- WEBAPI_STRING_FIELD</li> <li>- WEBAPI_BOOLEAN_FIELD</li> <li>- WEBAPI_ARRAY_FIELD</li> <li>- WEBAPI_OBJECT_FIELD</li> <li>- WEBAPI_BINARY_FIELD</li> <li>- WEBAPI_DATETIME_FIELD</li> <li>- WEBAPI_DATE_FIELD</li> </ul> |
| bRequired        | Boolean            | Determines if this field is required.                                                                                                                                                                                                                                                                                                                                              |
| nestedFields     | tFieldDefinition[] | This can be filled if there are objects or arrays nested inside of eachother. Is only used whenever the eFieldType is set to REST_ARRAY_FIELD or REST_OBJECT_FIELD.                                                                                                                                                                                                                |

#### 4.1.9 tParameterDefinition

This struct has all the information needed for additional parameters. These parameters are query or header parameters.

| tParameterDefinition |         |                                                                                                                                                                                                                                                                                                 |
|----------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Member name          | Type    | Description                                                                                                                                                                                                                                                                                     |
| sName                | String  | The name of the parameter.                                                                                                                                                                                                                                                                      |
| sDescription         | String  | The description what the parameter is about.                                                                                                                                                                                                                                                    |
| eParameterIn         | Integer | Determines if the parameter is a query or header parameter. The options for this are:<br>- WEBAPI_QUERY_PARAMETER<br>- WEBAPI_HEADER_PARAMETER                                                                                                                                                  |
| eParameterType       | Integer | The field type of the parameter. Can be one of the following:<br>- WEBAPI_INTEGER_FIELD<br>- WEBAPI_NUMBER_FIELD<br>- WEBAPI_STRING_FIELD<br>- WEBAPI_BOOLEAN_FIELD<br>- WEBAPI_ARRAY_FIELD<br>- WEBAPI_OBJECT_FIELD<br>- WEBAPI_BINARY_FIELD<br>- WEBAPI_DATETIME_FIELD<br>- WEBAPI_DATE_FIELD |
| bRequired            | Boolean | Determines if this parameter is required.                                                                                                                                                                                                                                                       |

## 4.2 Constants

The table below shows a list of all the constants used. It shows the name of the constant, the value and what it represents.

| Constants             |           |                                                                          |
|-----------------------|-----------|--------------------------------------------------------------------------|
| Name                  | Value     | Description                                                              |
| C_WEBAPI_GET          | "GET"     | The GET verb.                                                            |
| C_WEBAPI_POST         | "POST"    | The POST verb.                                                           |
| C_WEBAPI_PUT          | "PUT"     | The PUT verb.                                                            |
| C_WEBAPI_PATCH        | "PATCH"   | The PATCH verb.                                                          |
| C_WEBAPI_DELETE       | "DELETE"  | The DELETE verb.                                                         |
| C_WEBAPI_OPTIONS      | "OPTIONS" | The OPTIONS verb.                                                        |
| C_WEBAPI_OK           | 200       | Represents the 200 OK status code.                                       |
| C_WEBAPI_CREATED      | 201       | Represents the 201 CREATED status code.                                  |
| C_WEBAPI_NOCONTENT    | 204       | Represents the 204 NOCONTENT status code.                                |
| C_WEBAPI_NOTMODIFIED  | 304       | Represents the 304 NOTMODIFIED status code.                              |
| C_WEBAPI_BADREQUEST   | 400       | Represents the 400 BADREQUEST status code.                               |
| C_WEBAPI_UNAUTHORIZED | 401       | Represents the 401 UNAUTHORIZED status code.                             |
| C_WEBAPI_FORBIDDEN    | 403       | Represents the 403 FORBIDDEN status code.                                |
| C_WEBAPI_NOTFOUND     | 404       | Represents the 404 NOTFOUND status code.                                 |
| C_WEBAPI_NOTALLOWED   | 405       | Represents the 405 NOTALLOWED status code.                               |
| C_WEBAPI_ERROR_TYPE   |           | Link to the mozilla page containing the description of all status codes. |

## 4.3 Enums

This chapter has the information related to all the enums used inside of the framework.

### 4.3.1 Field types

This enum list represents all the available field types.

| Field types           |       |                   |
|-----------------------|-------|-------------------|
| Name                  | Value | Description       |
| WEBAPI_STRING_FIELD   | 0     | A string field.   |
| WEBAPI_NUMBER_FIELD   | 1     | A number field.   |
| WEBAPI_INTEGER_FIELD  | 2     | A integer field.  |
| WEBAPI_BOOLEAN_FIELD  | 3     | A Boolean field.  |
| WEBAPI_ARRAY_FIELD    | 4     | A array field.    |
| WEBAPI_OBJECT_FIELD   | 5     | A object field.   |
| WEBAPI_BINARY_FIELD   | 6     | A binary field.   |
| WEBAPI_DATETIME_FIELD | 7     | A datetime field. |
| WEBAPI_DATE_FIELD     | 8     | A date field.     |

### 4.3.2 Parameter types

This enum list represents all the types for query parameters.

| Parameter types         |       |                     |
|-------------------------|-------|---------------------|
| Name                    | Value | Description         |
| WEBAPI_QUERY_PARAMETER  | 0     | A query parameter.  |
| WEBAPI_HEADER_PARAMETER | 1     | A header parameter. |